



The 66th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Immunomodulatory Microbial Metabolites Protect Against Graft-Vs-Host Disease and Enhance Anti-Leukemia Responses in Mice and Patients Receiving Stem Cell Transplantation

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Background: The success of allogeneic hematopoietic stem cell transplantation (allo-HSCT) hangs on a delicate balance between preventing graft-versus-host disease (GvHD) while simultaneously promoting the elimination of residual malignant cells by an allogeneic immune reaction of donor T cells (graft-versus-leukemia (GvL)). Microbiome diversity and composition have been associated with clinical outcomes after allo-HSCT, including overall survival and incidence of GvHD [1-3]. Still, little is known about whether and how the microbiome modulates GvL. Our group recently established an "Immuno-Modulatory Metabolite Risk Index (IMM-RI)" comprised of microbiota-derived immuno-modulatory metabolites (IMMs) that were predictive of overall survival and relapse in patients receiving allo-HSCT [4].

The effect of IMMs has been partially attributed to type-I interferon (IFN-I) responses, which have been associated with positive or negative outcomes on GvHD, depending on the timing of IFN-I activation. We hypothesized microbiota-derived metabolites engage IFN-I signaling in epithelial and immune cells, poising them to modulate GvHD and GvL. To address the underlying cellular and molecular mechanisms, we screened for IMMs in allo-HSCT patients and tested them as treatment in patient *ex vivo* and murine *in vivo* models.

Methods Stool samples from patients receiving allo-HSCT were obtained at Rechts der Isar, Munich, and University Hospital Regensburg per IRB-approved study protocols (German Clinical Trials Register (DRKS00034175)). We quantified levels of microbiota-derived metabolites by mass spectrometry. IMMs detected in patients were tested as treatment in murine and human immune cells, intestinal crypt-derived organoids, and organoid/T cell co-culture assays. To test for the requirement for IFN-I signaling, we stimulated WT or IFN-I-signaling-impaired organoids (MAVS^{-/-}, STING^{gt/gt}, IFNAR^{-/-}) or used small molecule inhibitors and analyzed them regarding growth performance, intestinal stem cell (ISC) numbers, and by scRNAseq. IMMs were used to treat intestinal damage and major mismatch GvHD mouse models. Outcomes were assessed using a novel organoid recovery assay in addition to established read-outs (histology, barrier integrity, immune infiltrate). IMM treatment was also performed in germ-free or antibiotic-induced dysbiosis models to test for the requirement of microbiome signals.

Results: We demonstrate that IMMs correlate with improved overall survival and reduced relapse rates in allo-HSCT patients. In preclinical models, IMM treatment showed dose-dependent effects. Low-dose IMM moderately enhanced graft-vs-leukemia responses and prevented graft-vs.-host disease by protecting the intestinal barrier and promoting intestinal regeneration, while high doses promoted T cell activation and risked hyperacute GvHD. IMM treatment remained effective despite broad-spectrum antibiotic treatment, but not in germ-free mice. Mechanistically, IMM promoted mTORC1-dependent activation and proliferation of ISC, with the innate immune receptor STING required to mitigate metabolic stress and maintain an undifferentiated stem cell state. Consequently, the protective effect of IMMs was abrogated in STING-deficient mice.

Conclusions: Our findings underscore the potential of IMM's epithelial- and immune-modulating properties, paving the way for the design of microbiome-based therapies that could significantly improve outcomes in allo-HSCT and potentially other cancer immunotherapies.

References

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